

GVASYUGI, Russian Federation

WMO: 318010

Lat: 47.6704N

Lon: 136.1836E

Elev: 180

StdP: 99.18

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.6	-34.6	-39.4	0.1	-36.4	-37.4	0.1	-34.6	5.5	-18.5	5.0	-17.5	0.1	50	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.2	21.6	28.4	20.7	26.8	20.0	23.2	27.7	22.1	26.3	21.2	25.2	1.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.8	25.4	20.7	15.7	24.4	19.8	14.9	23.5	70.2	27.9	66.1	26.4	62.7	25.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.5	3.4	-39.7	33.1	2.4	1.5	-41.4	34.2	-42.7	35.1	-44.1	35.9	-45.8	37.0		
			-39.7	25.6	2.3	1.4	-41.4	26.6	-42.8	27.4	-44.1	28.3	-45.8	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-22.0	-17.9	-8.1	2.7	11.2	16.9	20.8	19.5	13.0	3.4	-9.5	-20.2	(6)	
(7)		DBStd	15.90	5.08	5.32	6.05	4.48	3.68	3.31	2.52	2.75	3.88	4.49	6.69	5.50	(7)	
(8)		HDD10.0	4334	992	782	560	222	31	1	0	0	16	207	586	937	(8)	
(9)		HDD18.3	6526	1251	1016	818	469	223	65	8	19	163	462	836	1196	(9)	
(10)		CDD10.0	1016	0	0	0	3	67	208	336	293	106	4	0	0	(10)	
(11)		CDD18.3	166	0	0	0	0	1	22	85	54	4	0	0	0	(11)	
(12)		CDH23.3	1468	0	0	0	6	84	326	677	342	33	1	0	0	(12)	
(13)		CDH26.7	377	0	0	0	1	16	93	204	61	3	0	0	0	(13)	
(14)	Wind	WSAvg	1.0	0.7	1.0	1.3	1.3	0.9	0.8	0.8	0.9	1.1	1.1	0.8	(14)		
(15)	Precipitation	PrecAvg	866	19	18	32	48	78	98	164	161	109	65	42	31	(15)	
(16)		PrecMax	1381	34	36	69	115	181	176	326	299	313	120	67	83	(16)	
(17)		PrecMin	607	5	1	3	12	26	38	45	49	49	12	9	11	(17)	
(18)		PrecStd	182	8	9	20	26	33	39	66	60	54	27	18	15	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.8	1.2	10.5	23.4	28.5	32.0	32.6	30.4	26.5	20.8	8.8	-2.0	(19)	
(20)			MCWB	-7.0	-1.9	3.8	11.5	16.0	20.6	22.7	22.8	19.3	12.8	4.9	-3.6	(20)	
(21)		2%	DB	-9.1	-2.6	7.0	18.8	25.7	29.3	30.8	28.3	23.8	16.3	4.9	-5.4	(21)	
(22)			MCWB	-10.1	-4.9	1.8	9.3	14.7	20.0	22.5	21.8	17.6	10.1	1.8	-6.5	(22)	
(23)		5%	DB	-11.4	-5.4	4.3	14.7	23.2	27.0	29.0	26.8	21.8	13.3	2.2	-8.5	(23)	
(24)			MCWB	-12.2	-7.0	0.1	7.1	13.2	18.7	21.5	21.1	16.7	8.5	0.3	-9.4	(24)	
(25)		10%	DB	-13.4	-7.8	1.9	11.6	20.4	24.8	27.2	25.3	19.7	10.7	0.2	-11.2	(25)	
(26)			MCWB	-14.1	-9.3	-1.0	5.7	12.0	17.8	21.1	20.2	15.5	7.3	-1.1	-11.9	(26)	
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.1	-1.1	4.2	12.6	17.5	23.5	24.8	24.2	20.9	13.3	5.6	-3.2	(27)
(28)				MCDWB	-6.1	1.2	9.3	22.4	25.6	28.4	29.9	28.2	25.1	18.8	8.6	-2.1	(28)
(29)			2%	WB	-10.1	-4.7	2.3	10.0	15.7	20.9	23.5	22.9	18.9	11.1	2.4	-6.5	(29)
(30)				MCDWB	-9.3	-2.8	6.0	16.9	23.1	27.2	28.6	26.6	22.4	14.7	4.4	-5.5	(30)
(31)	5%		WB	-12.2	-7.0	0.6	7.7	14.3	19.6	22.5	21.9	17.4	9.5	0.5	-9.4	(31)	
(32)			MCDWB	-11.4	-5.3	3.5	14.0	20.9	25.2	27.3	25.4	20.5	12.5	2.0	-8.6	(32)	
(33)	10%		WB	-14.1	-9.3	-0.9	5.9	13.1	18.6	21.6	21.0	16.1	7.7	-1.2	-11.9	(33)	
(34)			MCDWB	-13.4	-7.8	1.9	11.1	18.4	23.4	25.9	24.4	19.0	10.3	0.2	-11.2	(34)	
(35)	Mean Daily Temperature Range		5% DB	MDBR	14.6	17.4	15.5	13.3	14.6	12.8	11.0	9.9	11.9	12.1	11.6	13.0	(35)
(36)				MCDBR	15.3	18.3	16.7	20.6	21.0	17.8	15.1	12.5	14.9	17.3	12.8	13.4	(36)
(37)				MCWBR	14.2	15.9	11.7	11.6	10.5	9.0	7.5	6.7	8.6	10.7	9.7	12.2	(37)
(38)			5% WB	MCDBR	14.8	17.6	14.5	19.2	17.4	15.0	13.0	10.8	12.5	15.3	12.0	12.7	(38)
(39)		MCWBR		13.7	15.5	10.2	11.2	9.3	8.2	7.3	6.4	7.8	10.2	9.2	11.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.351	0.426	0.531	0.559	0.522	0.496	0.454	0.417	0.430	0.383	0.316	(40)		
(41)		taud	2.062	1.904	1.768	1.729	1.774	1.962	2.089	2.213	2.221	2.014	2.008	2.022	(41)		
(42)		Ebn at Noon	742	766	762	723	723	757	772	790	781	675	616	653	(42)		
(43)		Edh at Noon	93	137	186	214	213	177	154	131	118	123	100	86	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.92	3.10	4.26	4.45	5.07	5.35	5.09	4.47	3.70	2.48	1.62	1.43	(44)		
(45)		RadStd	0.10	0.13	0.34	0.45	0.42	0.69	0.46	0.31	0.30	0.22	0.15	0.08	(45)		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page